

The Model as Judge

Potential applications of LLM-as-a-Judge

A practical workshop on where model-based evaluation helps, how to validate it, and when a human still needs to decide.

A SIMPLE JUDGE LOOP

PROMPT + CONTEXT

What did we ask? What evidence is available?

+ TWO CANDIDATE ANSWERS

ANSWER A

Clear, cites the source

ANSWER B

Longer, adds unsupported detail

JUDGE OUTPUT

A wins on groundedness / B needs review

A judge gives us a signal to inspect - not a new source of truth.

From opportunity to a trustworthy judge

01

Why judge?

Open-ended quality is hard to measure with exact-match tests.

02

Judge patterns

Single-answer grading, pairwise comparison, references.

03

Where it helps

RAG, release evaluation, agent traces, response quality.

04

How to specify it

A clear rubric and a structured, evidence-linked result.

05

How to trust it

Human agreement, bias checks, monitoring, escalation.

06

What to pilot

Start narrow, measure disagreements, then decide.

Quality is more than exact match

For open-ended answers, there may be many good ways to respond - and many subtle ways to fail.

A string comparison misses whether an answer followed the instruction, used the right evidence, covered the request, or stayed within policy.

A judge can turn those qualitative expectations into repeatable evaluation signals.

01

Instruction fit

Did it do what the user asked?

02

Groundedness

Are important claims supported?

03

Completeness

Did it address every material part?

04

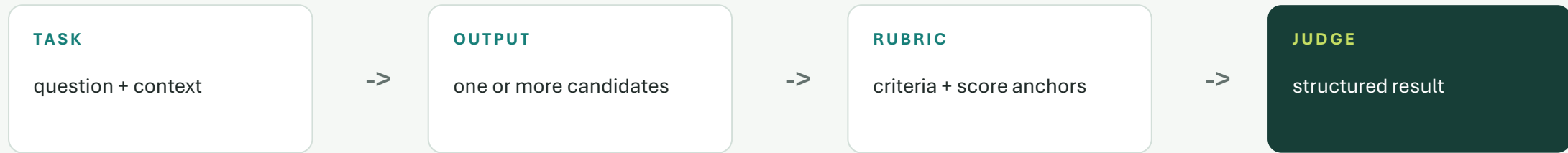
Safety & tone

Was it appropriate for the context?

One overall score can hide which quality dimension regressed.

What do we mean by LLM-as-a-Judge?

An LLM evaluates an output against a stated rubric - sometimes with a reference or source evidence - and returns a score, comparison, or verdict.



LLM-AS-A-JUDGE

An evaluation pattern. A generative model can apply a natural-language rubric and return a concise explanation.

SYSTEM ONE MODEL

A model class for bounded typed decisions. JEV / CLM can score or select; they do not generate an explanation. Related, not synonymous.

Three ways to ask a judge

01

Single-answer grading

Grade one candidate on each rubric dimension.

Answer -> score 1-5 + evidence

Best for: release checks, quality trends, rubric gates.

02

Pairwise comparison

Show two answers to the same task; choose A, B, or tie.

A vs B -> winner / tie

Best for: prompt or model selection, ranking candidates.

03

Reference-guided

Compare an answer with a trusted reference or supplied sources.

Answer + reference -> supported?

Best for: factuality, extraction, grounding.

Pairwise preference is relative; an A/B win does not prove either answer meets an absolute quality bar.

Where judge models could add leverage

Prompt / model selection

Compare versions on the same examples; surface wins, ties, and regressions.

Response quality

Check helpfulness, instruction fit, completeness, and tone against a rubric.

RAG groundedness

Check whether retrieved passages support the answer's material claims.

Summaries & extraction

Assess coverage, omissions, and unsupported additions when exact-match is too brittle.

Agent trace review

Evaluate tool choice, arguments, progress toward goal, and use of observations.

Safety triage

Flag possible policy violations for review; keep enforcement and authorization in code.

Highest potential: repeated judgments with an explicit rubric and an affordable path to human review.

RAG: judge the pipeline, not just the prose

01

Context relevance

Do retrieved passages address the question?

02

Answer faithfulness

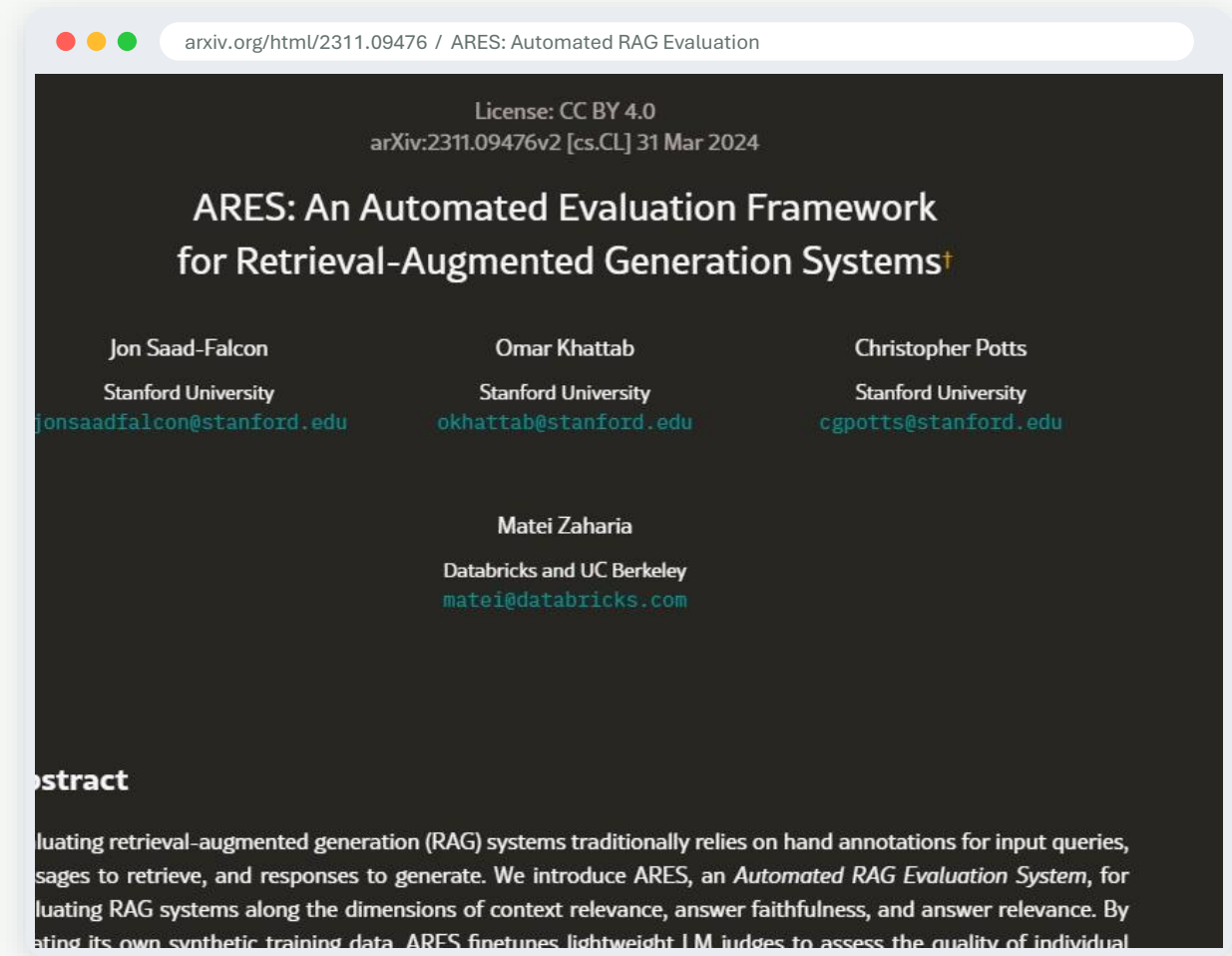
Are the answer's claims supported by those passages?

03

Answer relevance

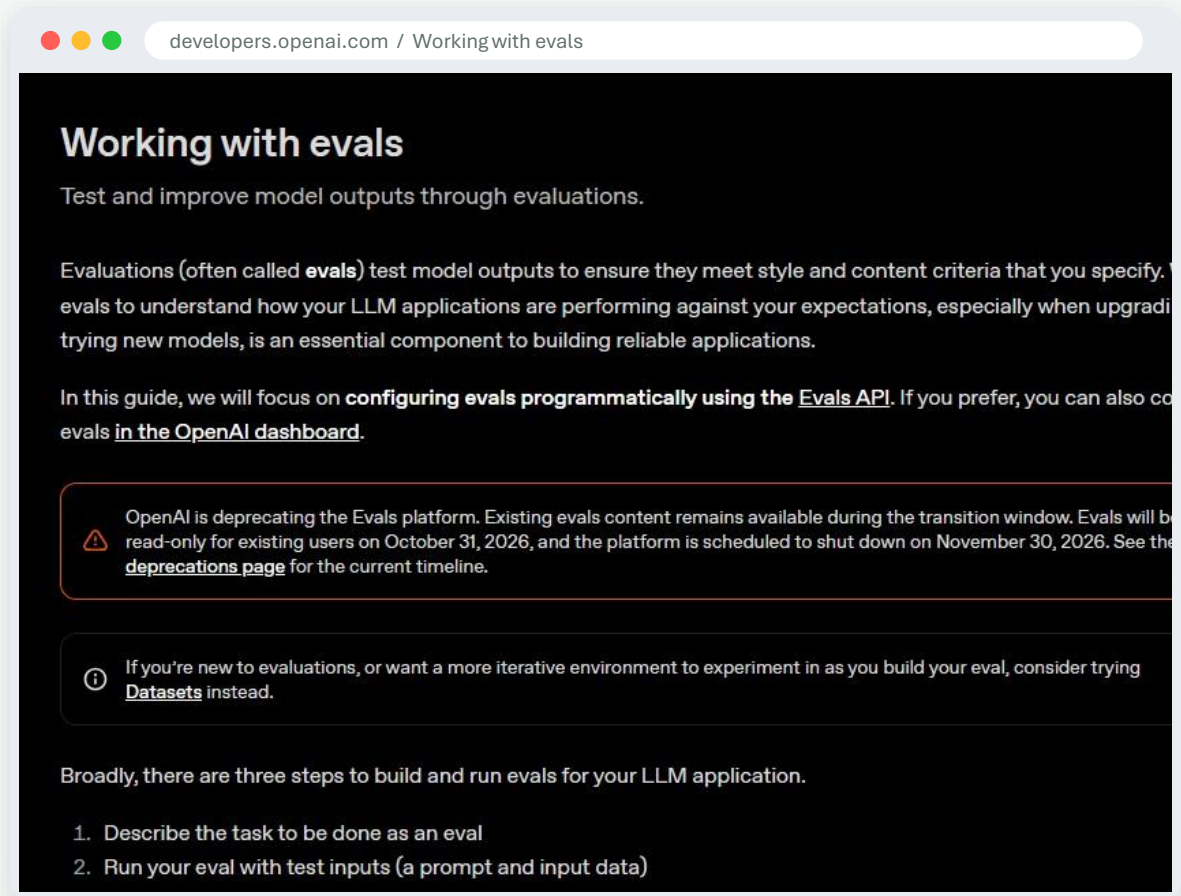
Does the answer actually resolve the question?

Keep the three scores separate. A grounded answer can still be irrelevant; a relevant answer can still be unsupported.



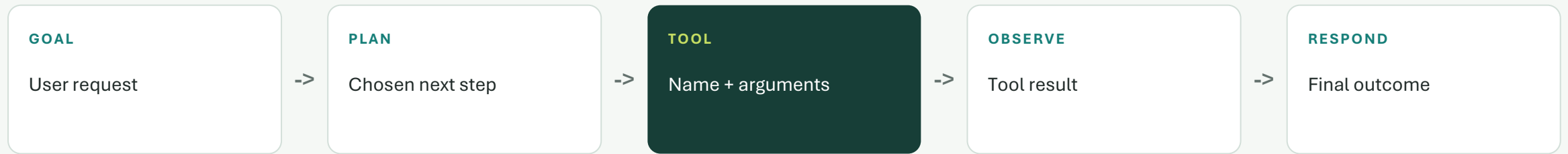
Prompt and model releases: catch regressions before users do

01	Freeze examples	Representative inputs + human labels or acceptance criteria.
02	Run candidates	Same inputs across prompt, model, and config versions.
03	Judge by rubric	Score dimensions separately; retain evidence.
04	Compare slices	Find where quality improved, slipped, or became uncertain.
05	Inspect & decide	Review disagreements before setting a release gate.



Evaluation concepts shown; verify current platform and API status before adopting a tool.

Agent evaluation: score the trajectory, not only the final answer



Goal progress

Did each step move toward the requested outcome?

Tool choice

Was this tool appropriate for the task and state?

Argument quality

Were parameters supported by context or evidence?

Observation use

Did the agent incorporate the tool result accurately?

A judge can assess behavior. Deterministic policy and permissions must still control tool execution.

Where a judge fits - and where it does not

STRONG FIT

- Open-ended output with multiple acceptable answers
- Repeated evaluation where human review is expensive
- Clear criteria such as relevance, completeness, or tone
- Low-risk triage with a review path

JUDGE + HUMAN

- Safety or policy edge cases
- Close model comparisons and ambiguous outputs
- High-impact decisions or uncertain evidence
- New domains before judge calibration

USE AN EXACT CHECK / EXPERT

- Arithmetic or exact schema validation
- Authorization, access, or business rules
- Legal / medical conclusions without expert oversight
- Cases with a reliable deterministic oracle

Rule of thumb: automate the measurement loop before automating the decision itself.

A useful rubric reads like a test, not a vibe

Ask for observable evidence, one dimension at a time, with anchors for every score.

CRITERION	Faithfulness
QUESTION	Are material claims supported by the supplied sources?
0	A material claim contradicts or lacks support.
1	Some claims are supported; one material claim is unclear or unsupported.
2	All material claims are supported by the supplied sources.

Avoid "Is this a good answer?"

RUBRIC DESIGN CHECK

- 1 Specific** Define what counts as a pass or a miss.
- 2 Separated** Keep faithfulness, coverage, and style distinct.
- 3 Evidence-based** Ask for the exact claim and supporting source.
- 4 Calibratable** Include examples and a review path.
- 5 Stable** Version the rubric; test it like code.

Make the verdict structured, reviewable, and safe

A judge result should explain what to inspect - not ask downstream code to parse a paragraph.

label	pass review fail
scores	per-criterion bounded scale
evidence	claim + source or output span
summary	one-sentence reason
abstain	true when evidence is insufficient

EXAMPLE RESULT / GROUNDEDNESS

```
{
  "label": "review",
  "scores": {
    "faithfulness": 1,
    "completeness": 2
  },
  "evidence": [{
    "claim": "ships in two days",
    "source_id": "S2",
    "supported": false
  }],
  "summary": "One delivery claim lacks support.",
  "abstain": false
}
```

Validate the schema in code. Treat thresholds as routing signals; never let a judge grant permissions or bypass policy.

Validate the judge against people before relying on it

01

Build

Representative cases, edge cases, and known failures.

02

Label

Human judgments with a rubric; record disagreement.

03

Compare

Run the judge blind on a held-out set.

04

Analyze

Agreement, per-class errors, disagreement examples.

05

Calibrate

Choose thresholds for the cost of false pass / fail.

06

Monitor

Sample live cases; re-check after any change.

MEASURE

human agreement

false-pass / false-fail

per-slice gaps

abstain rate

cost + latency

A judge score is not calibrated confidence unless probability estimates are tested against outcomes.

Known judge biases have practical countermeasures

POSITION BIAS

A/B ordering changes the winner.

Swap order; inconsistent verdicts become ties or review.

VERBOSITY BIAS

Longer can look better even when repetitive.

Score the rubric, not response length or flourish.

IDENTITY / STYLE

Judge may prefer familiar or its own model styles.

Blind model names; rotate independent judges on a sample.

REASONING LIMITS

Judge may accept incorrect math or be misled by a candidate.

Use references, deterministic checks, and expert review.

Bias controls reduce known failure modes; they do not make a judge objective.

Evidence is promising - and task-specific

80%+

Human agreement

MT-Bench / Chatbot Arena reported GPT-4 agreement above 80% on studied preference tasks - around the human-human agreement in that setup.

0.514

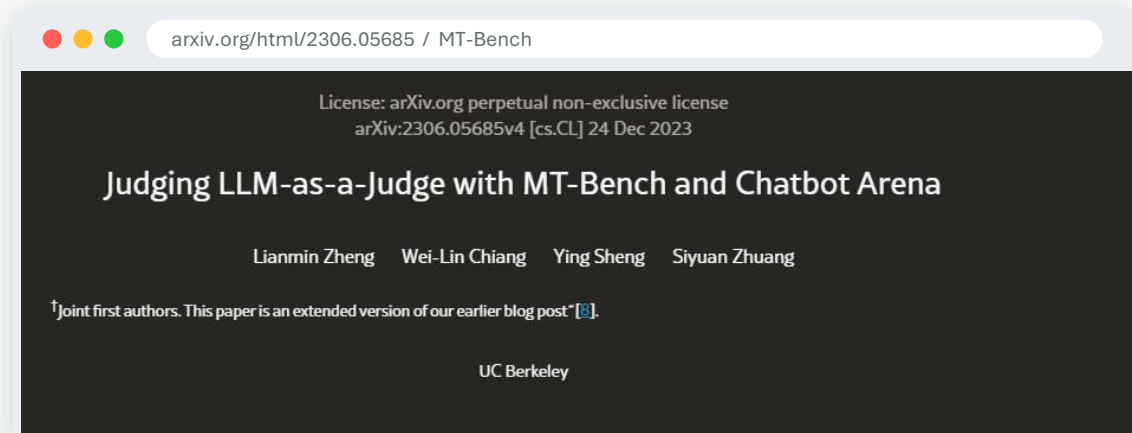
Summarization correlation

G-Eval reported Spearman correlation with human ratings on its summarization experiment. Not a universal judge score.

ARES

RAG evaluation

Component-level judges plus a few hundred human labels and prediction-powered inference across studied tasks.



LLM judge and System One are complementary

GENERATIVE JUDGE

Can interpret a richer rubric

Useful when evaluation needs multiple criteria, a short rationale, or evidence explanations. Output can be flexible - and therefore needs validation.

Trade-off: more generation, variable latency and cost.

SYSTEM ONE MODEL

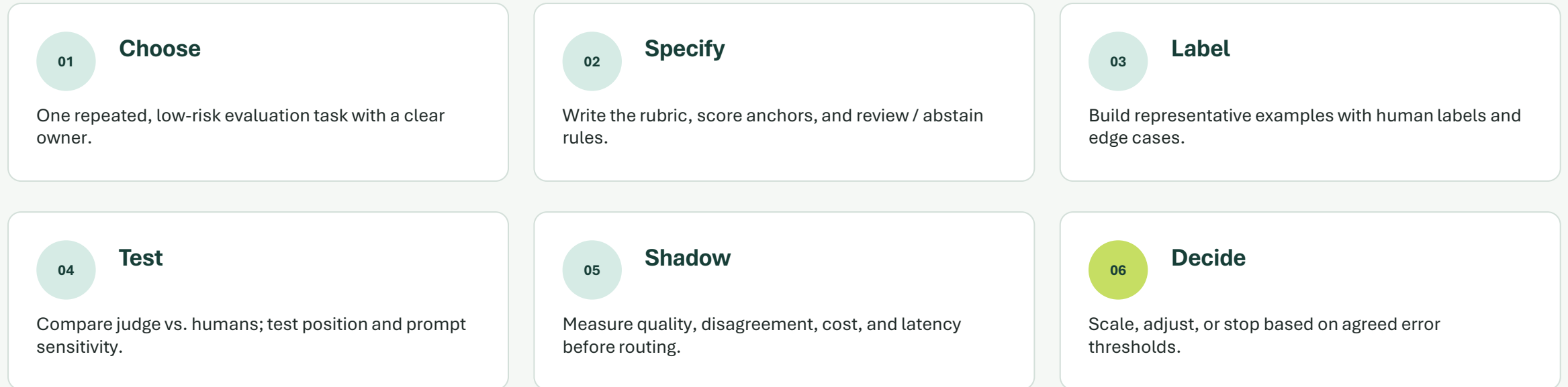
Fast, bounded decisions

JEV is TypeSafe's early-access System One product; CLM is an open-weight model. Both return typed choices, scores, or probabilities - not explanations.

Trade-off: answer space and questions must be defined up front.

Possible pattern: bounded model for first-pass scoring -> generative judge or human review for evidence and ambiguous cases.

Start with a narrow question and a human reference



GO / NO-GO

human agreement

false-pass rate

worst-case slices

review volume

cost / latency

Do not automate a high-impact action just because average judge agreement looks good.

Potential in practice

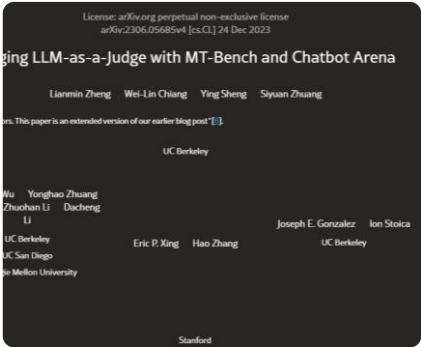
The best first use is not the most impressive one. It is the one we can measure honestly.

PROMPTS FOR THE ROOM

- 1 Which outputs do people review repeatedly today?
- 2 What would a good answer look like in observable terms?
- 3 Which error is costly enough to require human review?
- 4 What evidence would change our mind about a pilot?

Questions & discussion

Research and implementation references



MT-Bench / Chatbot Arena

Zheng et al. 2023 - agreement, judge types, and bias study.

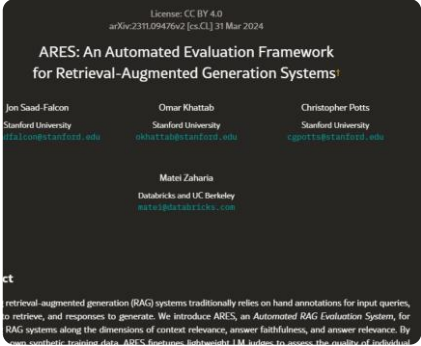
arxiv.org/abs/2306.05685



G-Eval

Liu et al. 2023 - rubric-based evaluation for NLG.

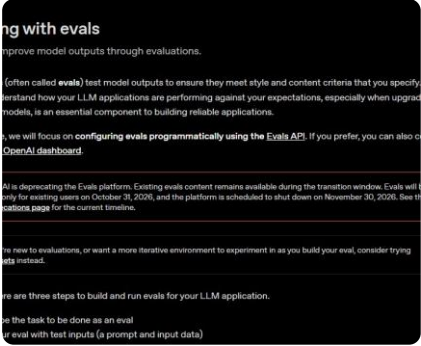
arxiv.org/abs/2303.16634



ARES

Saad-Falcon et al. 2024 - automated evaluation of RAG systems.

arxiv.org/abs/2311.09476



Evaluation guide

OpenAI docs - evaluation data, graders, runs, and analysis.

developers.openai.com/api/docs/guides/evals

System One context: TypeSafe JEV launch post, CLM model card, and Sanity glossary. Product claims remain vendor-reported.